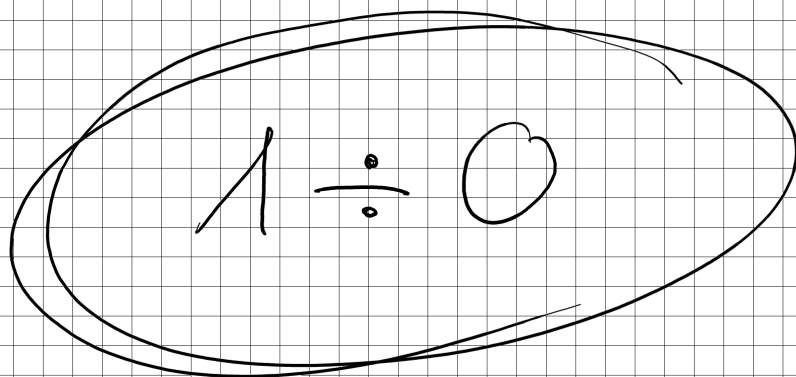




Dividing by zero

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Define: think $(\mathbb{Z} \times \mathbb{Z}) / \sim$

$$\mathbb{Q}_W = \{ \frac{a}{b} \mid a, b \in \mathbb{Z} \}$$

where: $\frac{a}{b} = \frac{c}{d} \Leftrightarrow$

$\exists x, y \in \mathbb{Z}^+$ such that

$$\begin{aligned} ax &= cy \\ bx &= dy \end{aligned}$$

Recall: $\mathbb{Q} = (\mathbb{Z} \times \mathbb{N}) / \sim$ where

$$(a, b) \sim (c, d) \Leftrightarrow ad = bc.$$

write $[(a, b)] = \frac{a}{b}$. (vollst. geordnet)

Question: Can we extend this to division by zero?

Standard attempt $\hat{\mathbb{Q}} = \mathbb{Q} \cup \{\infty\}$

- $0 \neq a \in \mathbb{Q}, \frac{a}{\infty} = \infty$

$$a \cdot \infty = \infty, a + \infty = \infty, \infty + \infty = \infty$$

- still undefined: $\frac{0}{0}, 0 \cdot \infty, \frac{\infty}{\infty}$



ex: $1 = \frac{1}{0} \cdot \frac{0}{1}$ brings problems $\rightarrow 1 = \frac{1}{0} \cdot \frac{0}{1} = \frac{1}{0} \cdot (0+0) = 1+1 = 2$

some observations

- we have normal fraction equality

assume $b, d \neq 0$ $\frac{a}{b} = \frac{c}{d} \Leftrightarrow ad = bc \Leftrightarrow \begin{matrix} x=d \\ y=b \end{matrix}$

$$\begin{aligned} ad &= bc \\ \updownarrow \\ ax &= cy \\ bx &= dy \\ bd &= bd \end{aligned}$$

- $a \neq 0, \frac{a}{0} = \frac{1}{0} \begin{cases} ax = 1 \cdot y \Leftrightarrow \begin{matrix} x=1 \\ y=a \end{matrix} \\ 0 \cdot x = 0 \cdot y \end{cases} \rightarrow \boxed{\infty := \frac{1}{0}}$

- $\frac{0}{0}$ is a "new element"

because $\frac{a}{b} = \frac{0}{0} = \begin{cases} a \cdot x = 0 \cdot y \\ b \cdot x = 0 \cdot y \end{cases} \Rightarrow a=b=0 \rightarrow \boxed{1 := \frac{0}{0}}$

types of elements $\boxed{a/b, b \neq 0} \quad \infty = 1/0 \quad 1 = 0/0$

addition: $a/b + c/d = (ad+bc)/bd$

multiplication: $a/b \cdot c/d = ac/bd$

• $1 + \infty = 1/1 + 1/0 = (1 \cdot 0 + 1 \cdot 1)/1 \cdot 0 = 1/0 = \infty$

$\boxed{a/b, b \neq 0}$
 $\boxed{a/b + \infty = \infty}$

• $1 + 1 = 1/1 + 0/0 = (1 \cdot 0 + 1 \cdot 0)/0 \cdot 1 = 0/0 = 1$

$\boxed{a/b, b \neq 0}$
 $\boxed{a/b + 1 = 1}$

• $\infty + \infty = 1/0 + 1/0 = (1 \cdot 0 + 0 \cdot 1)/0 \cdot 0 = 0/0 = 1$

Homework: • $\infty + 1 = 1$

• $1 + 1 = 1$

$\boxed{1/0 + 0/0 = 1 \cdot 0 + 0 \cdot 0 / 0 \cdot 0 = 0/0 = 1}$
 $\boxed{0/0 + 0/0 = 0 \cdot 0 + 0 \cdot 0 / 0 \cdot 0 = 0/0 = 1}$

Now multiplication $\boxed{a, b \neq 0}$ $\rightarrow a/b \cdot \infty = a/b \cdot 1/0 = a/0 = \infty$

$a/b \cdot 1 = a/b \cdot 0/0 = a/0 = 1$

$\infty \cdot \infty = 1/0 \cdot 1/0 = 1/0 = \infty$

$\infty \cdot 0 = 1/0 \cdot 0/1 = 0/0 = 1$

Distributive Rules?

$(2+3) \cdot \infty = 5 \cdot \infty = \infty$ $\neq$ $2 \cdot \infty + 3 \cdot \infty = \infty + \infty = 1$ ✗

$(2+3) \cdot 1 = 5 \cdot 1 = 1$ $\neq$ $2 \cdot 1 + 3 \cdot 1 = 1 + 1 = 1$ ✓

$(2+\infty) \cdot 3 = \infty \cdot 3 = \infty$, $2 \cdot 3 + \infty \cdot 3 = 6 + \infty = \infty$ ✓

1 $\rightarrow (2+3) \cdot \infty_{+ \infty \cdot 0} = 2 \cdot \infty + 3 \cdot \infty$

2 $\rightarrow (2+3) \cdot 1_{+ 0 \cdot 1} = 2 \cdot 1 + 3 \cdot 1$

3 $\rightarrow (2+\infty) \cdot 3_{+ 3 \cdot 0} = 2 \cdot 3 + \infty \cdot 3$

$\hookrightarrow$ new distributive rule: $(x+y) \cdot z + 0 \cdot z = x \cdot z + y \cdot z$

Definition A wheel is a set W together with commutative, associative, binary operations $+$ and $\cdot$ and a unary operation $/$ such that

• $1/x = x$ think $1/x = x$

• $1/x/y = 1/(xy)$ $1/x \cdot 1/y = 1/xy$

• $(x+y)z + 0 \cdot z = xz + yz$, $(x+yz)/y = x/y + z + 0y$

• $0 \cdot 0 = 0$ • $(x+0 \cdot y)z = xz + 0 \cdot y$

• $1/(x+0y) = 1/x + 0 \cdot y$ • $0/0 + x = 0/0$